

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1648

Warrensburg 100 kV
March 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1648. The Interconnection Customer proposes a 146.5 MW solar plus storage facility interconnecting at the Warrensburg 100 kV substation in the Illinois area. The request was received on January 12, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 553 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$325,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Talenenergy Martins Creek Llc Harrisburg 500/345kV XFMR	Coronet to Westwood Generation Llc 500 KV	106.7	1,584	5.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Talenenergy Martins Crk Llc Harrisburg 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$35,243,000	1% (\$325,000)
	Total allocated to project		\$325,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.